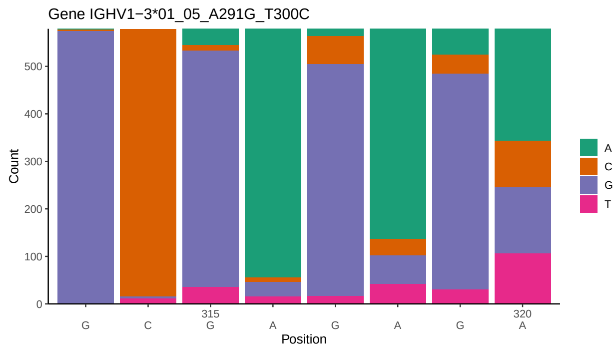
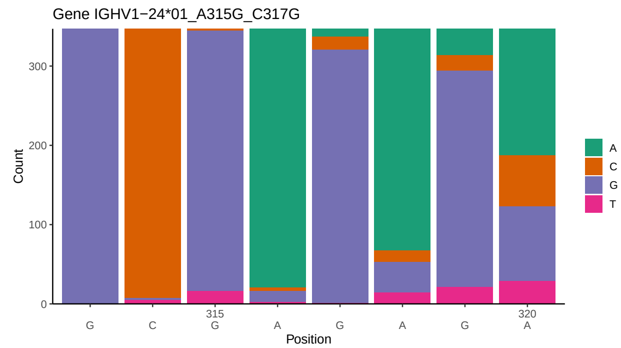
OGRDBstats Report

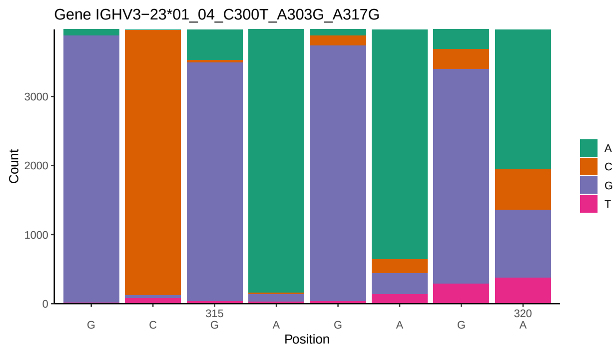
Contents

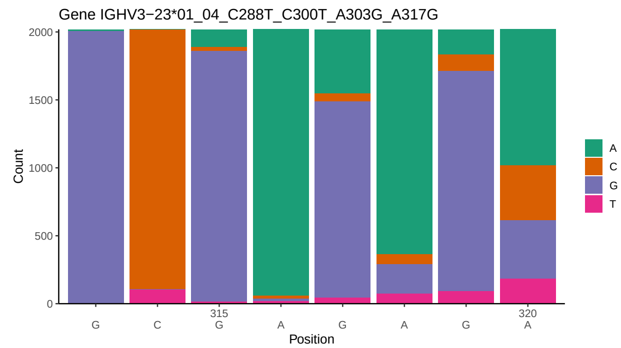
1 Novel sequence analysis

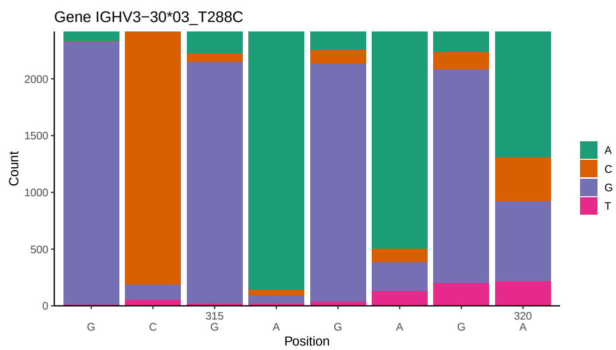
1.1 End-nucleotide composition

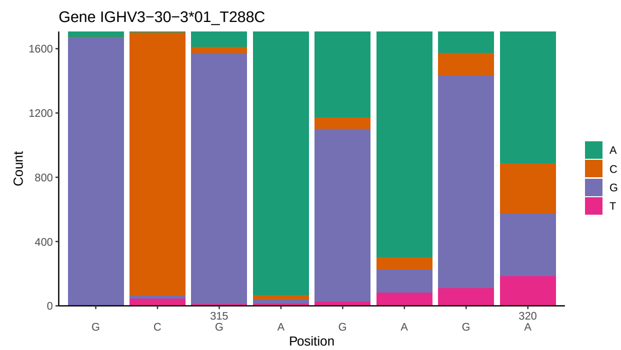


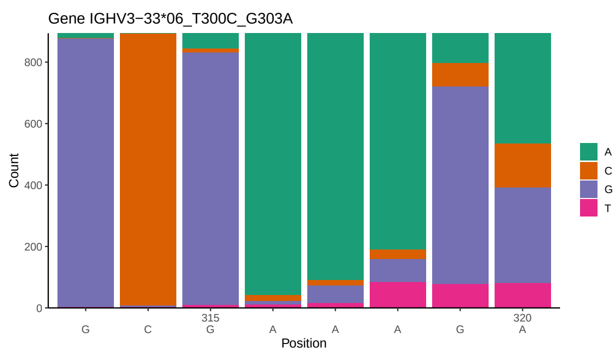


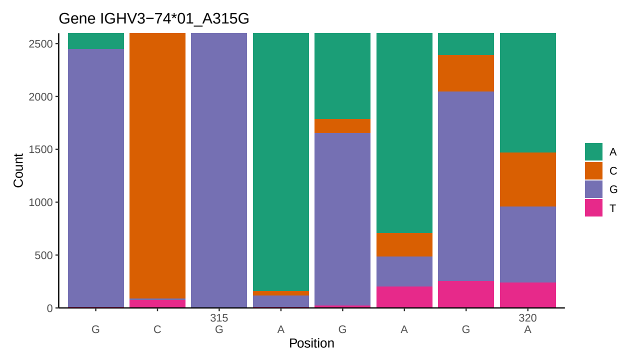


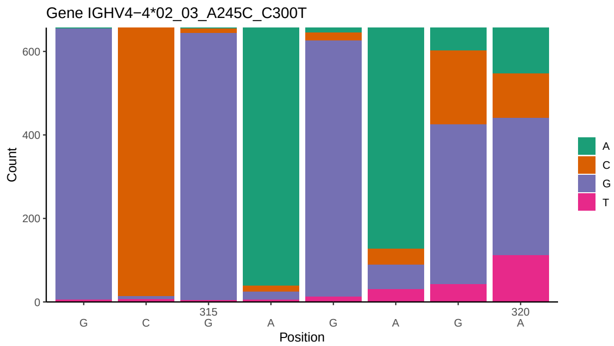


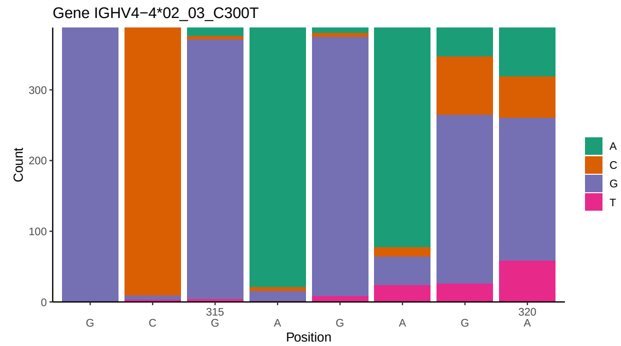


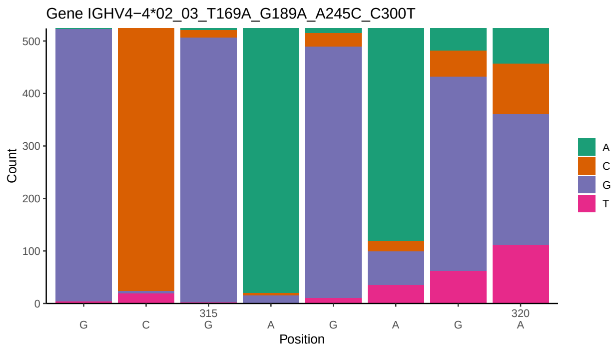


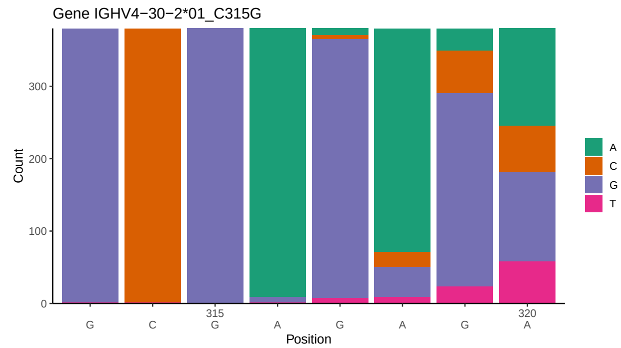


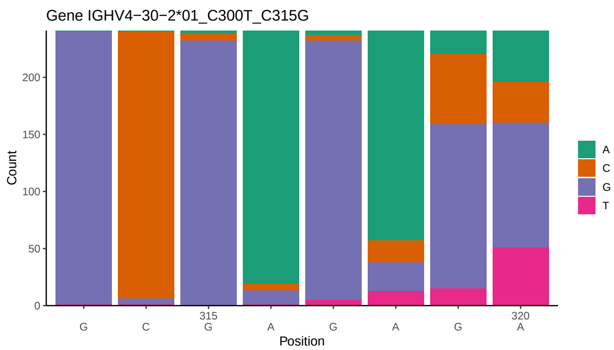


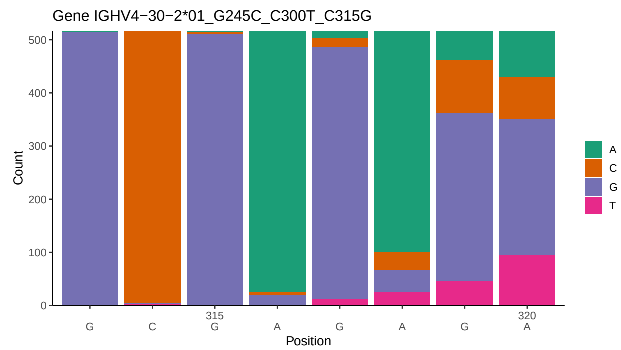

Gene IGHV4−4*02_03_A245C_C300T
Count
Position
A
C
G
T


Gene IGHV4−4*02_03_C300T
Count
Position
A
C
G
T


Gene IGHV4−4*02_03_T169A_G189A_A245C_C300T
Count
Position
A
C
G
T


Gene IGHV4−30−2*01_C315G
Count
Position
A
C
G
T


Gene IGHV4−30−2*01_C300T_C315G
Count
Position
A
C
G
T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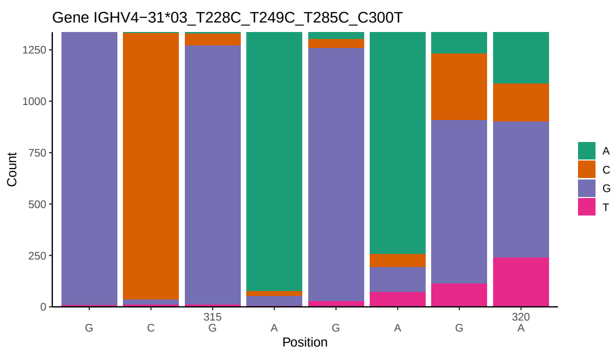
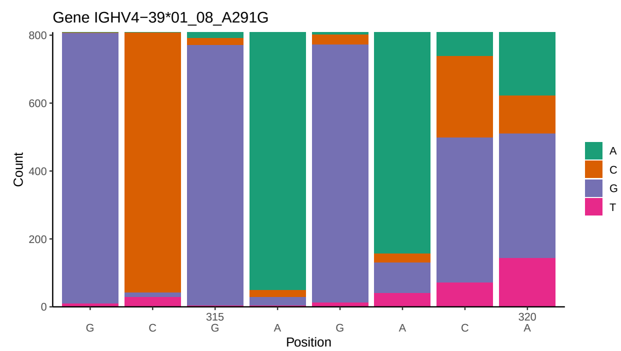
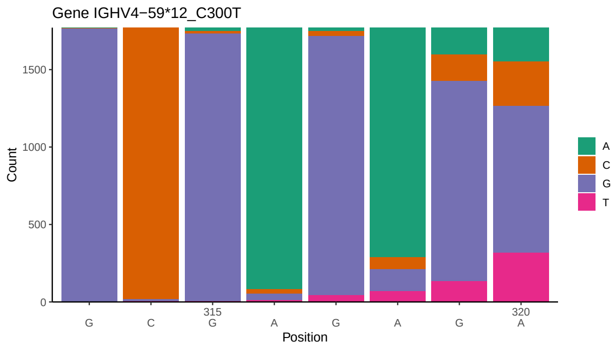
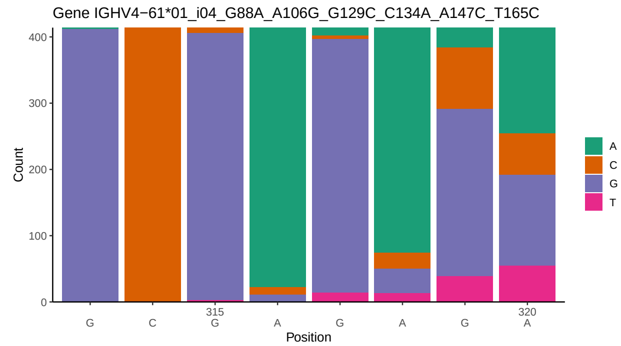
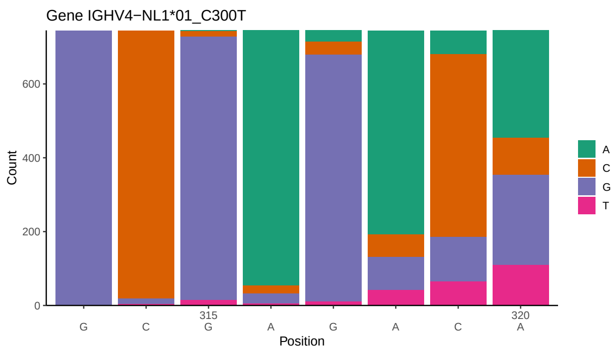
Count
Position
A
C
G
T


Gene IGHV4−31*03_T228C_T249C_T285C_C300T
Count
Position
A
C
G
T

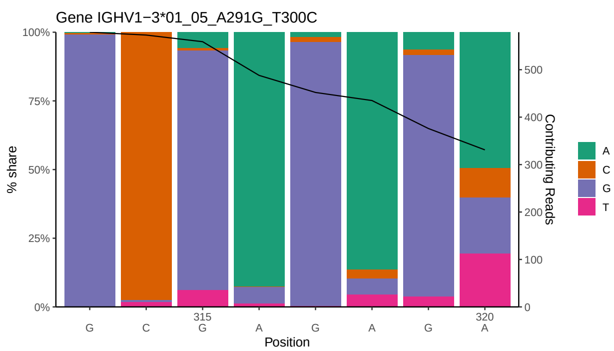

Gene IGHV4−39*01_08_A291G
Count
Position
A
C
G
T

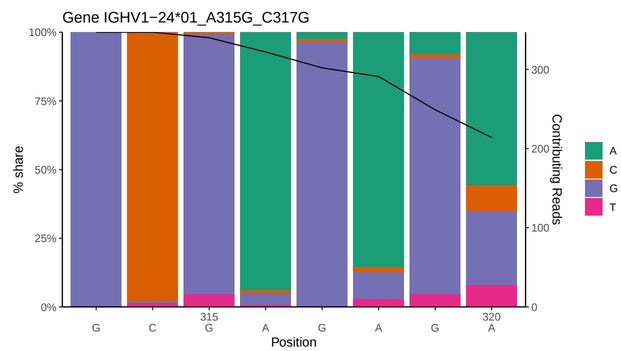


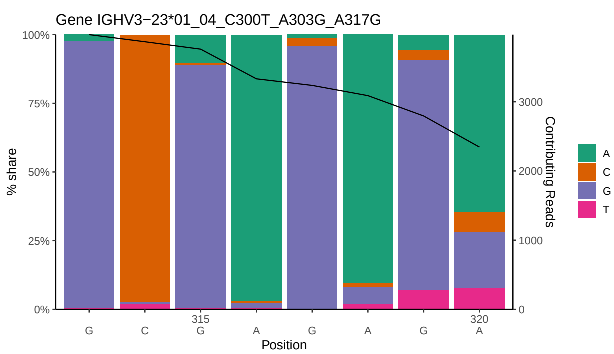


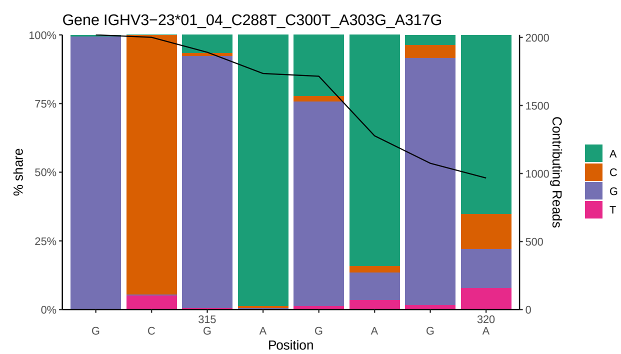


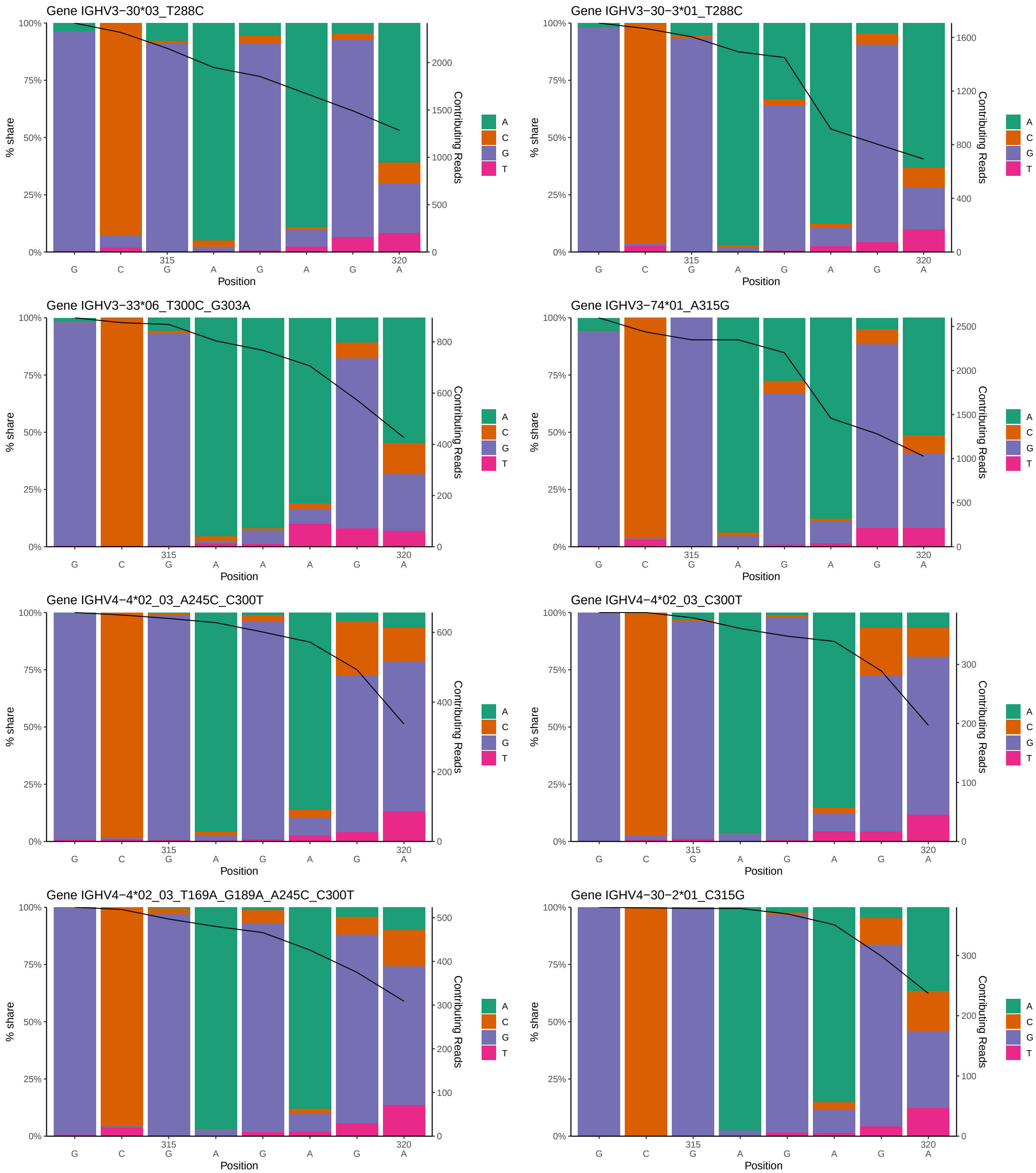
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

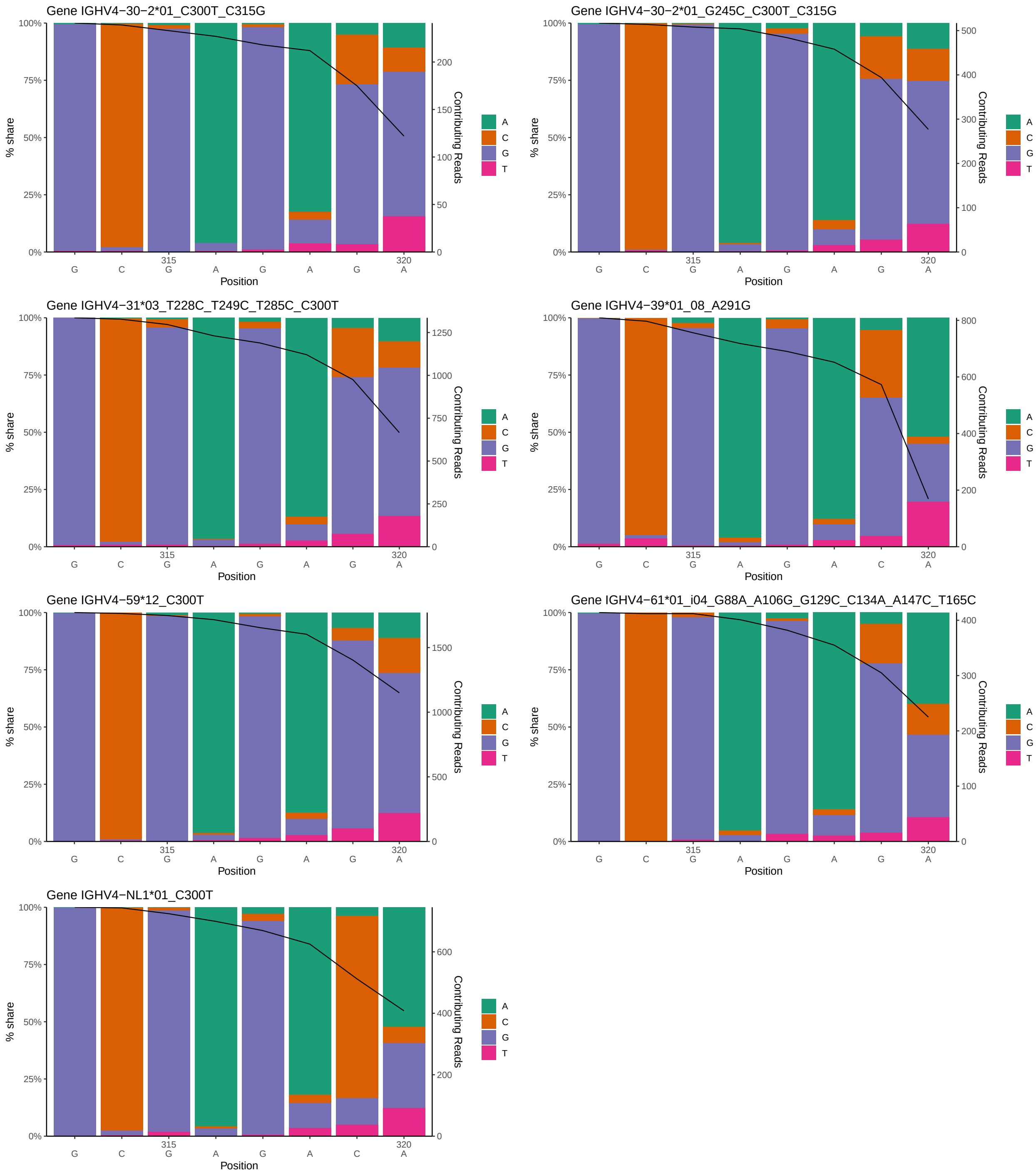




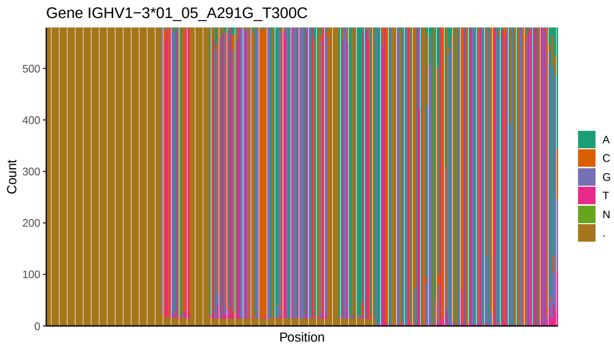


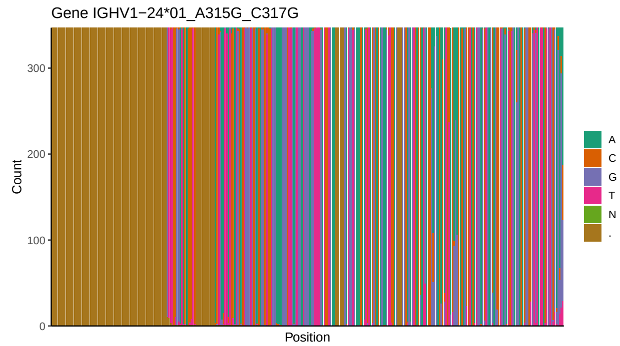


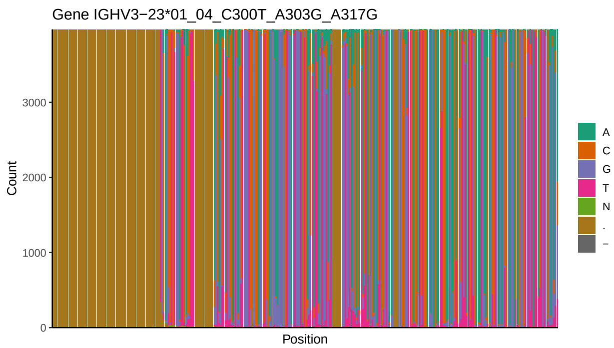

Gene IGHV3−30*03_T288C
Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_C300T
Gene IGHV4−4*02_03_T169A_G189A_A245C_C300T
Gene IGHV4−30−2*01_C315G
% share
Contributing Reads
Position
G C 315 G A G A G 320 A
A C G T

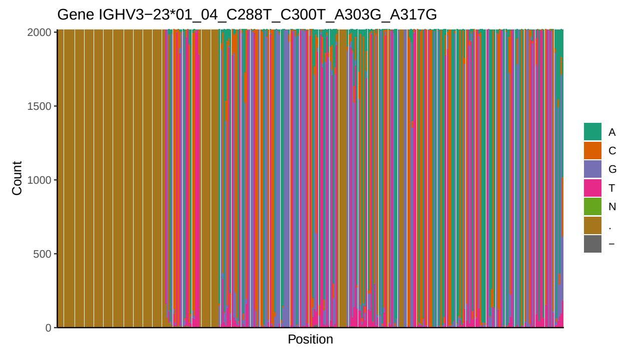

Gene IGHV4−30−2*01_C300T_C315G
Gene IGHV4−30−2*01_G245C_C300T_C315G
Gene IGHV4−31*03_T228C_T249C_T285C_C300T
Gene IGHV4−39*01_08_A291G
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_G88A_A106G_G129C_C134A_A147C_T165C
Gene IGHV4−NL1*01_C300T
% share
Contributing Reads
Position
A
C
G
T

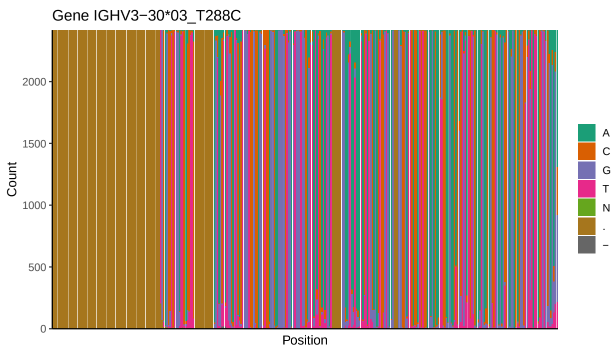
1.3 Whole-sequence composition of each assigned read

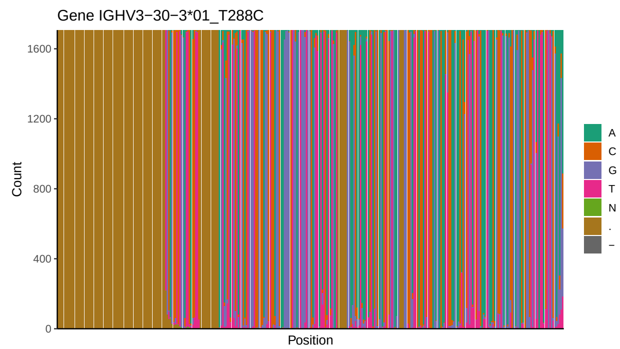


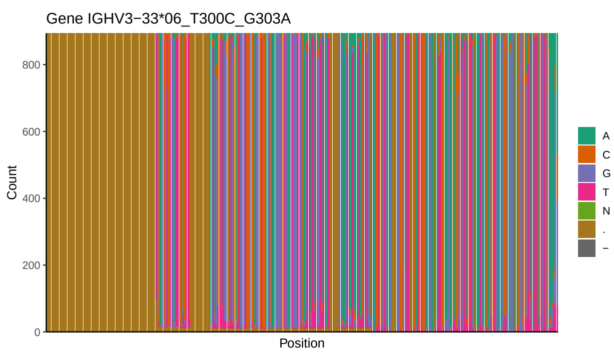


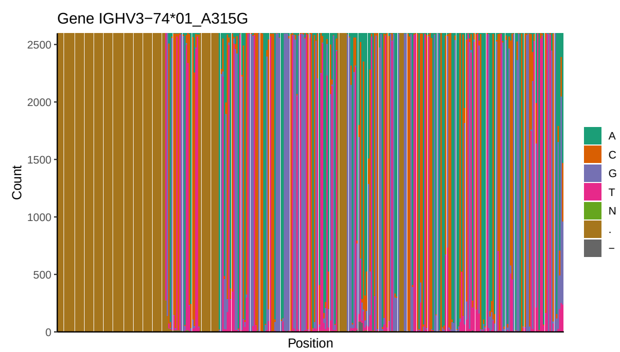




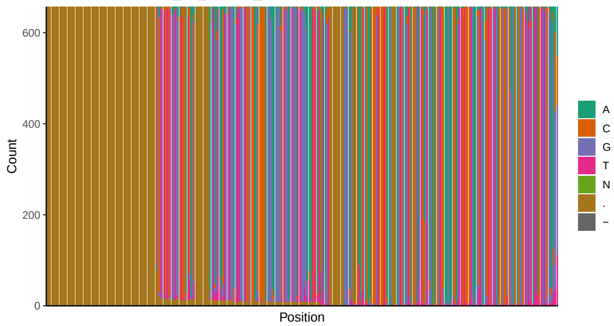






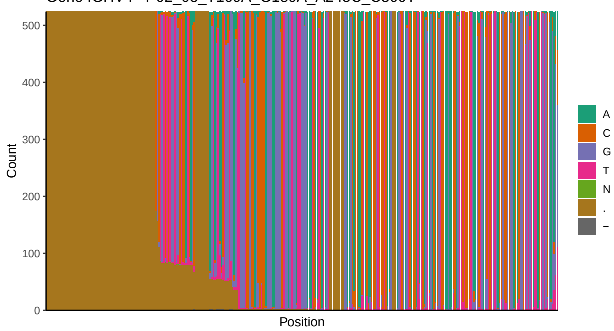


Gene IGHV4−4*02_03_A245C_C300T

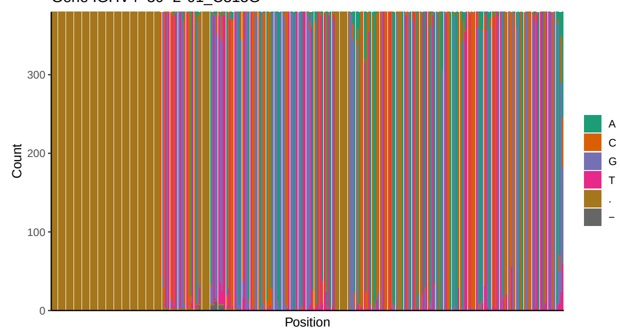


Gene IGHV4−4*02_03_C300T

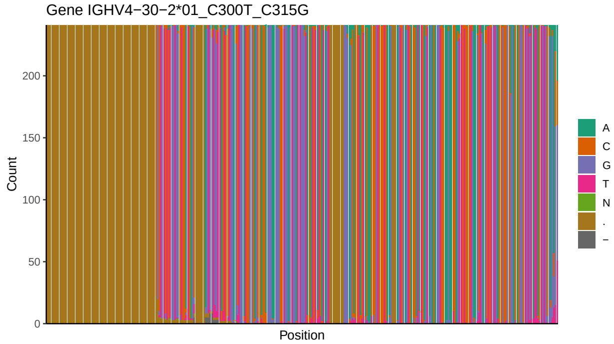
Gene IGHV4−4*02_03_T169A_G189A_A245C_C300T



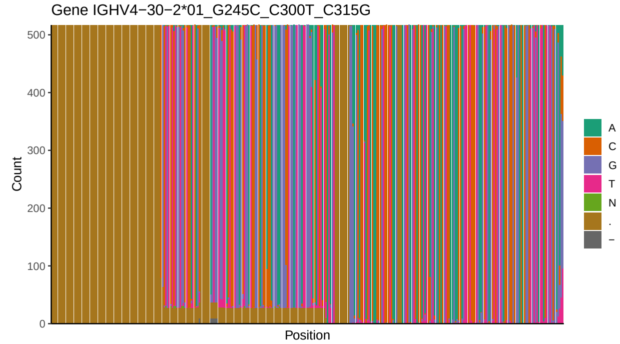
Gene IGHV4−30−2*01_C315G



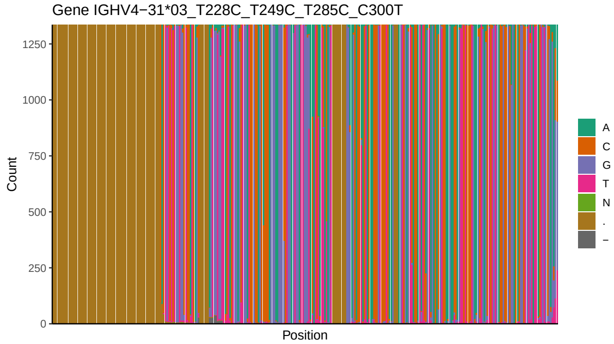
Gene IGHV4−30−2*01_C300T_C315G



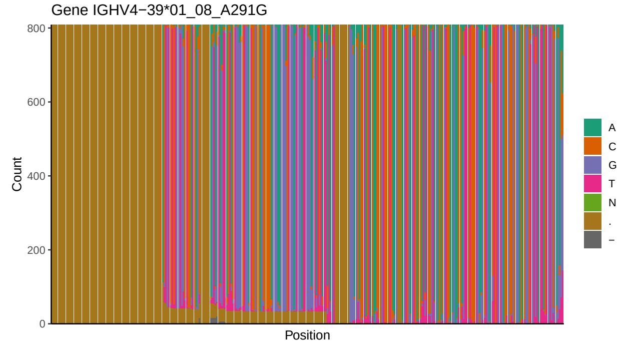
Gene IGHV4−30−2*01_G245C_C300T_C315G

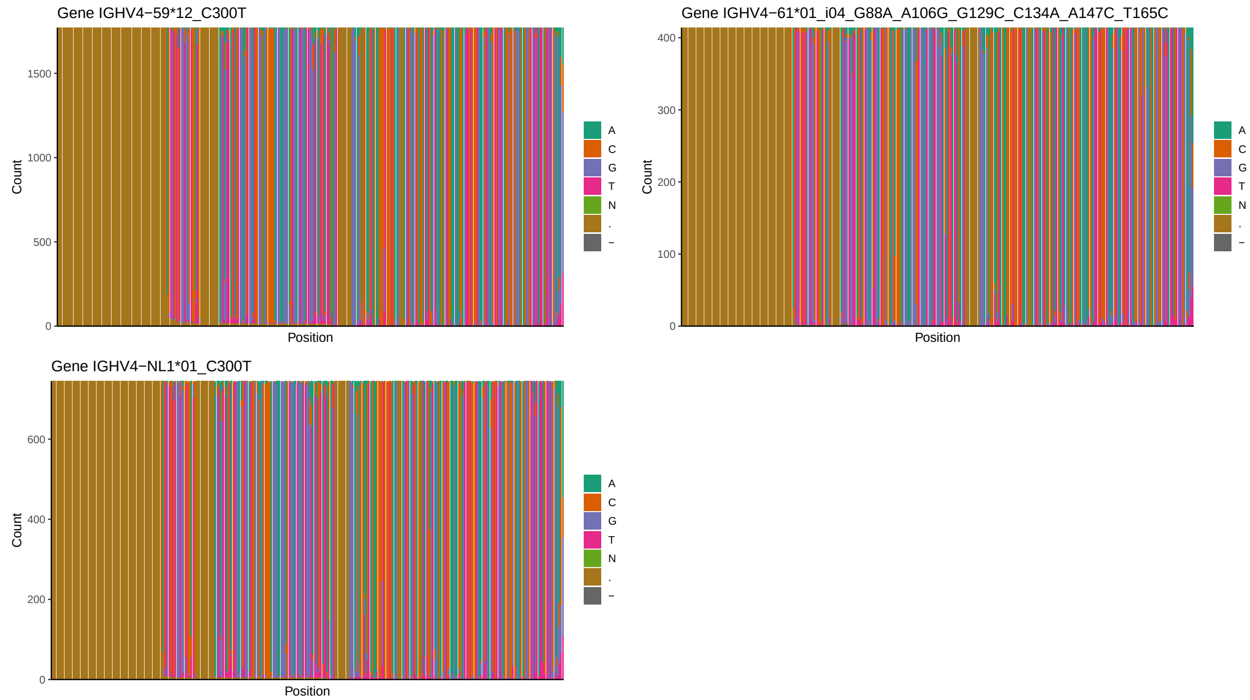


Gene IGHV4−31*03_T228C_T249C_T285C_C300T

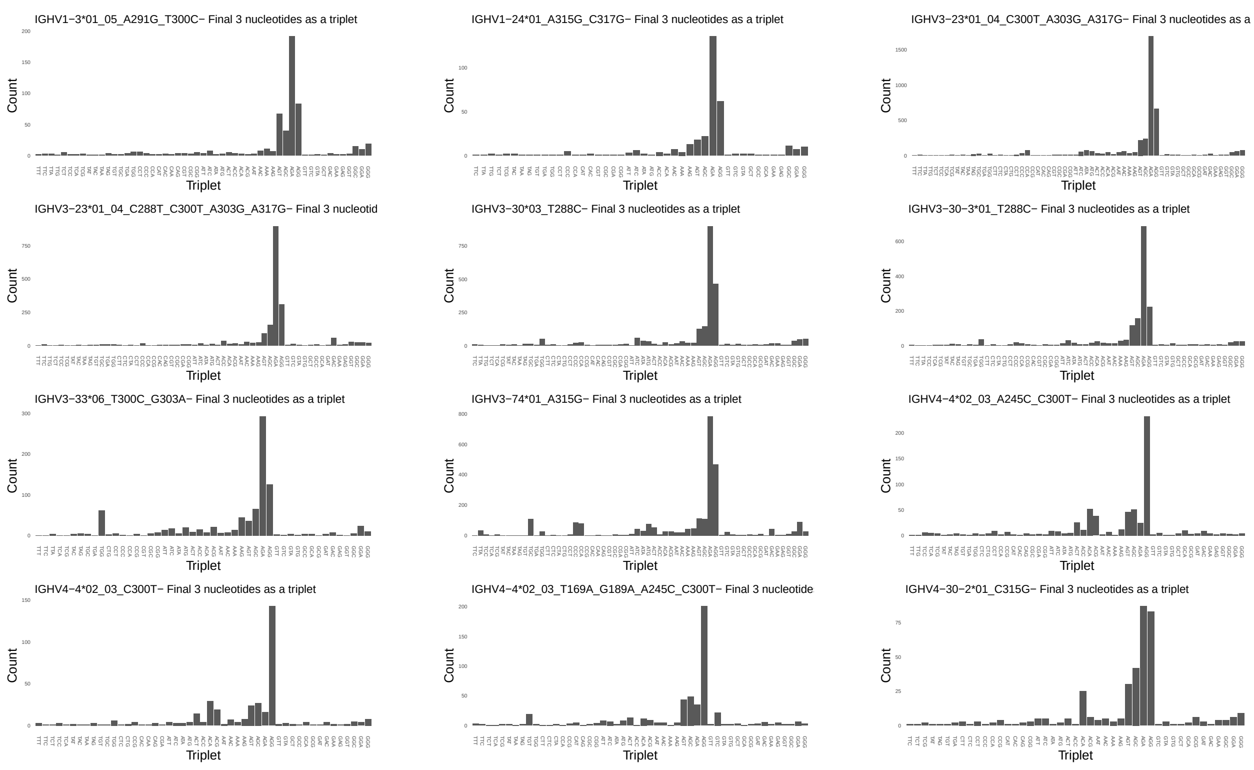


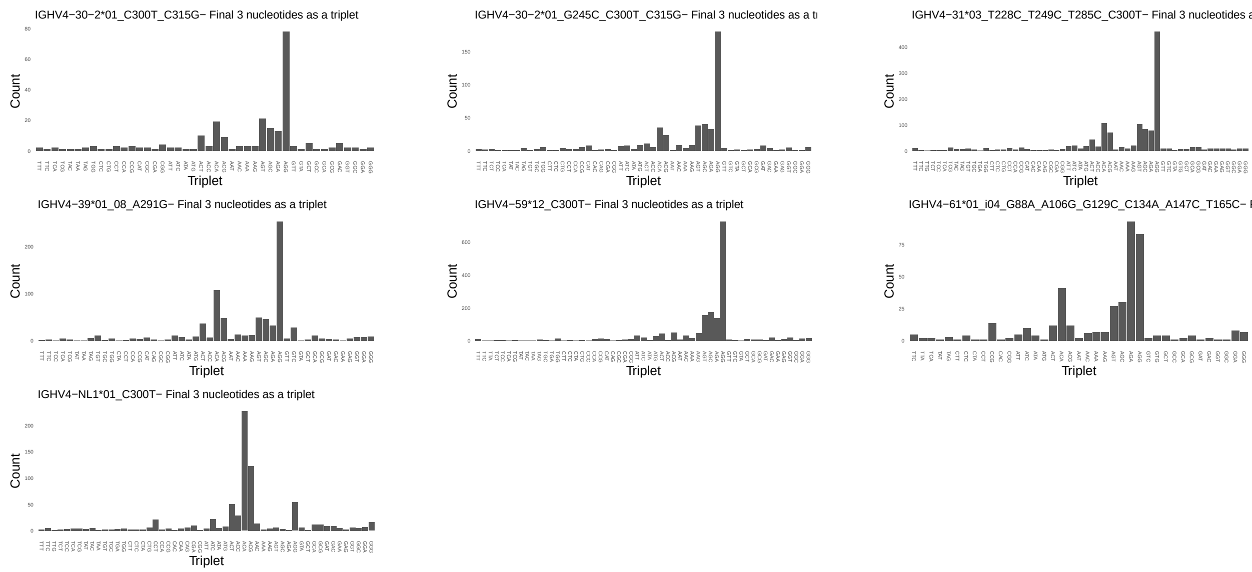
Gene IGHV4−39*01_08_A291G



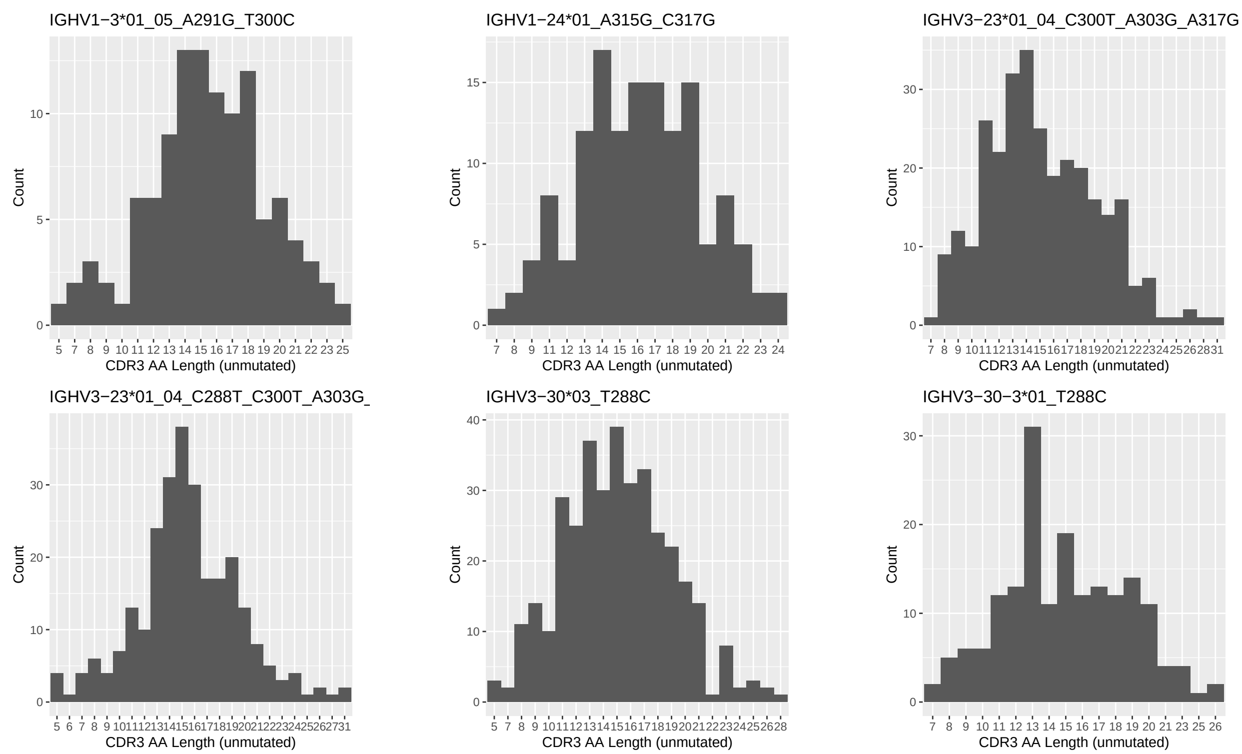


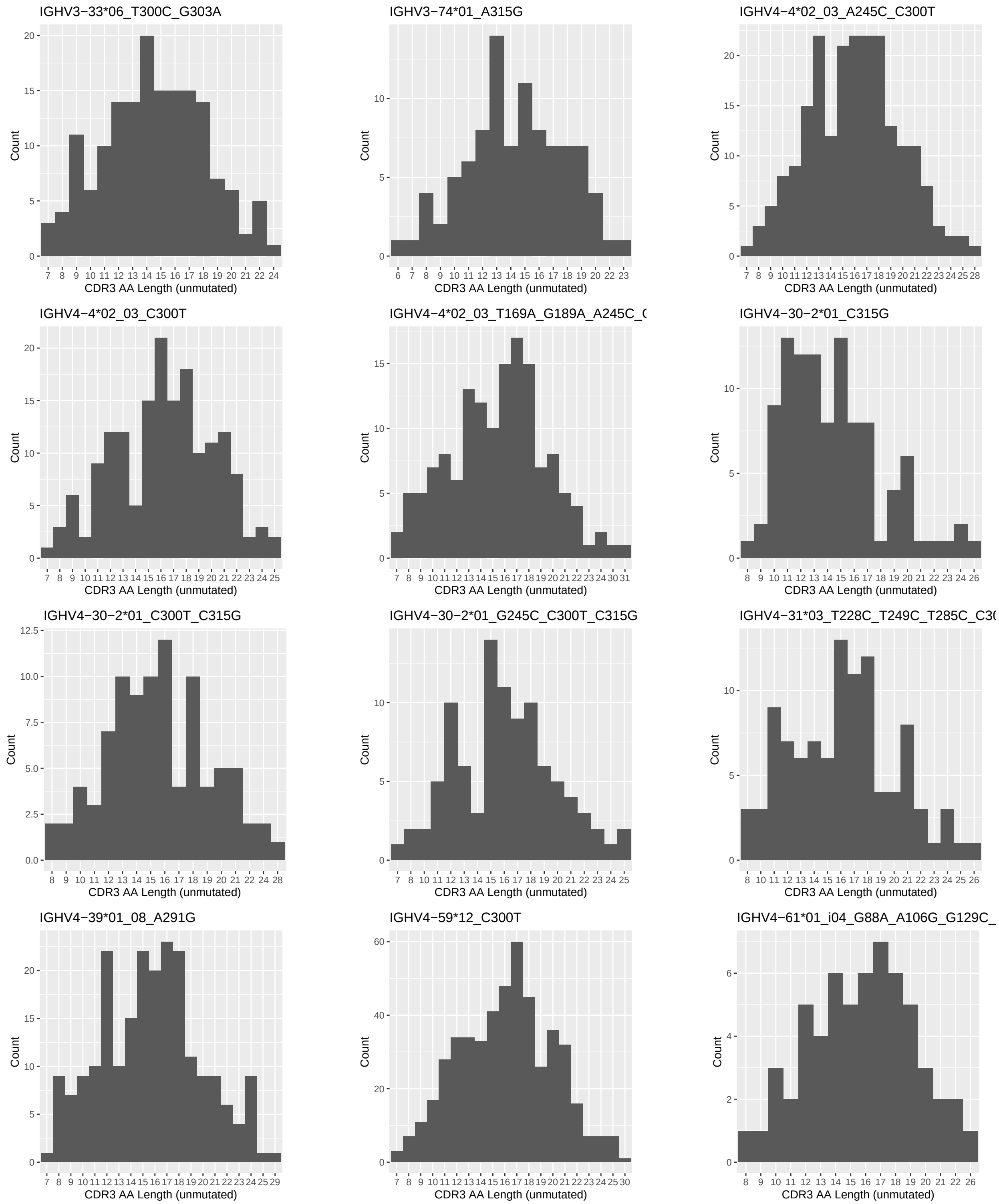
1.4 Final three nucleotides: frequency of each observed triplet

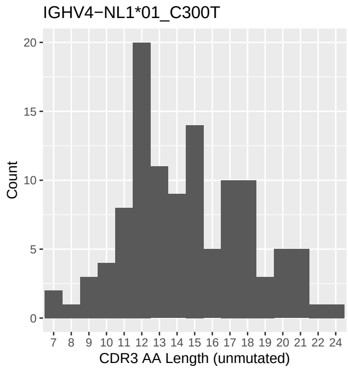




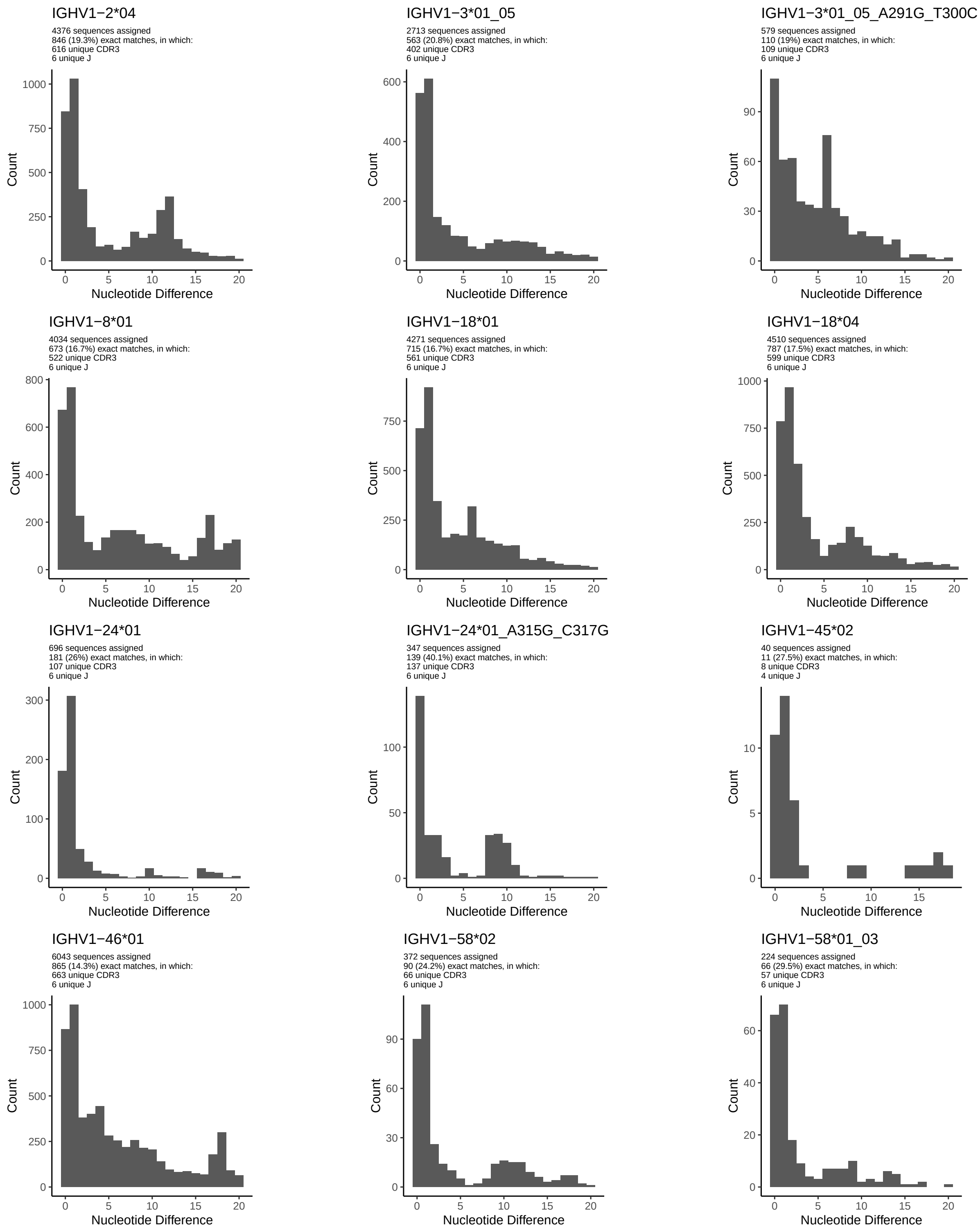
1.5 CDR3 length distribution, in assignments to novel alleles

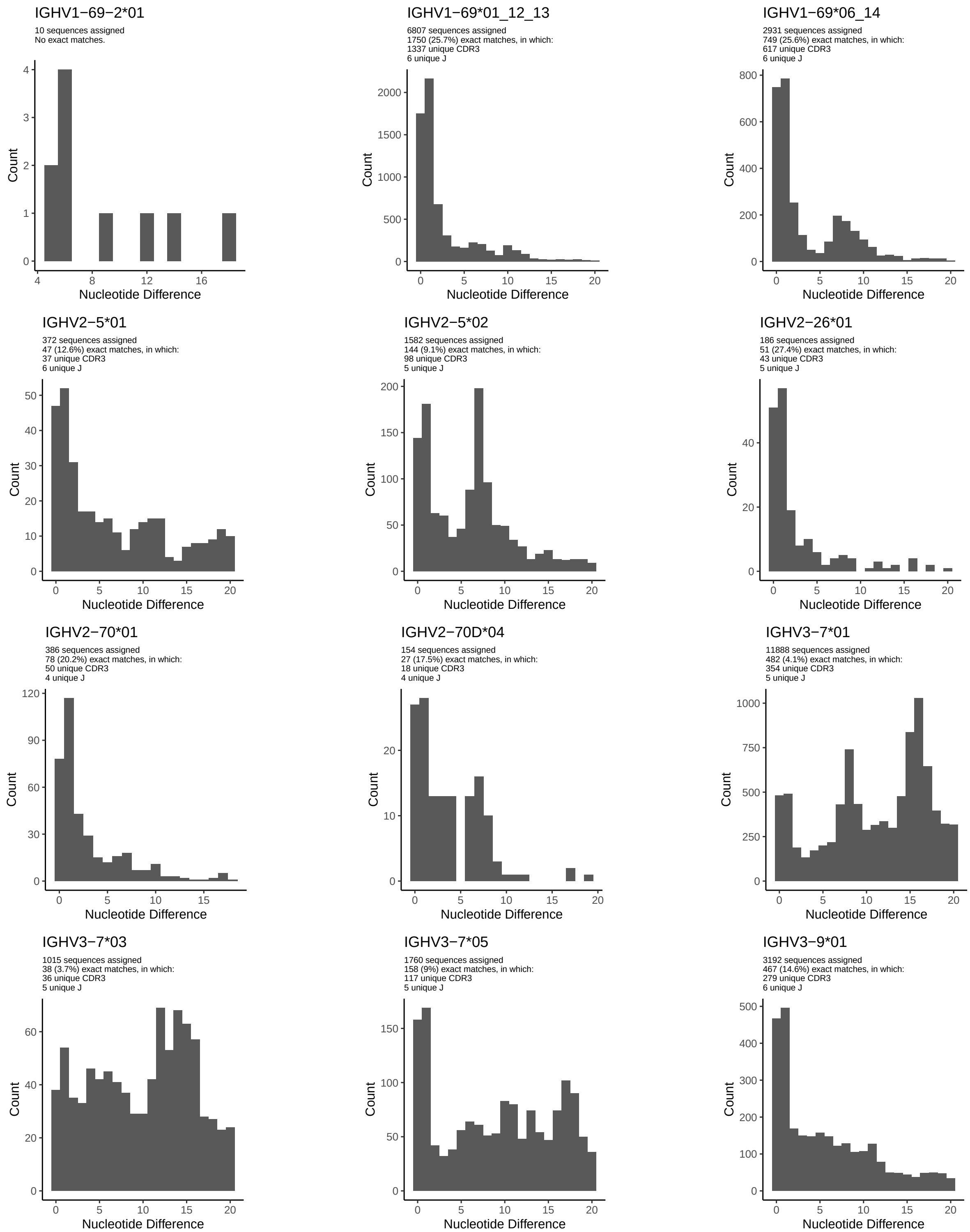



IGHV3−33*06_T300C_G303A
IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−4*02_03_C300T
IGHV4−4*02_03_T169A_G189A_A245C_C
IGHV4−30−2*01_C315G
IGHV4−30−2*01_C300T_C315G
IGHV4−30−2*01_G245C_C300T_C315G
IGHV4−31*03_T228C_T249C_T285C_C3(
IGHV4−39*01_08_A291G
IGHV4−59*12_C300T
IGHV4−61*01_i04_G88A_A106G_G129C_
Count
CDR3 AA Length (unmutated)

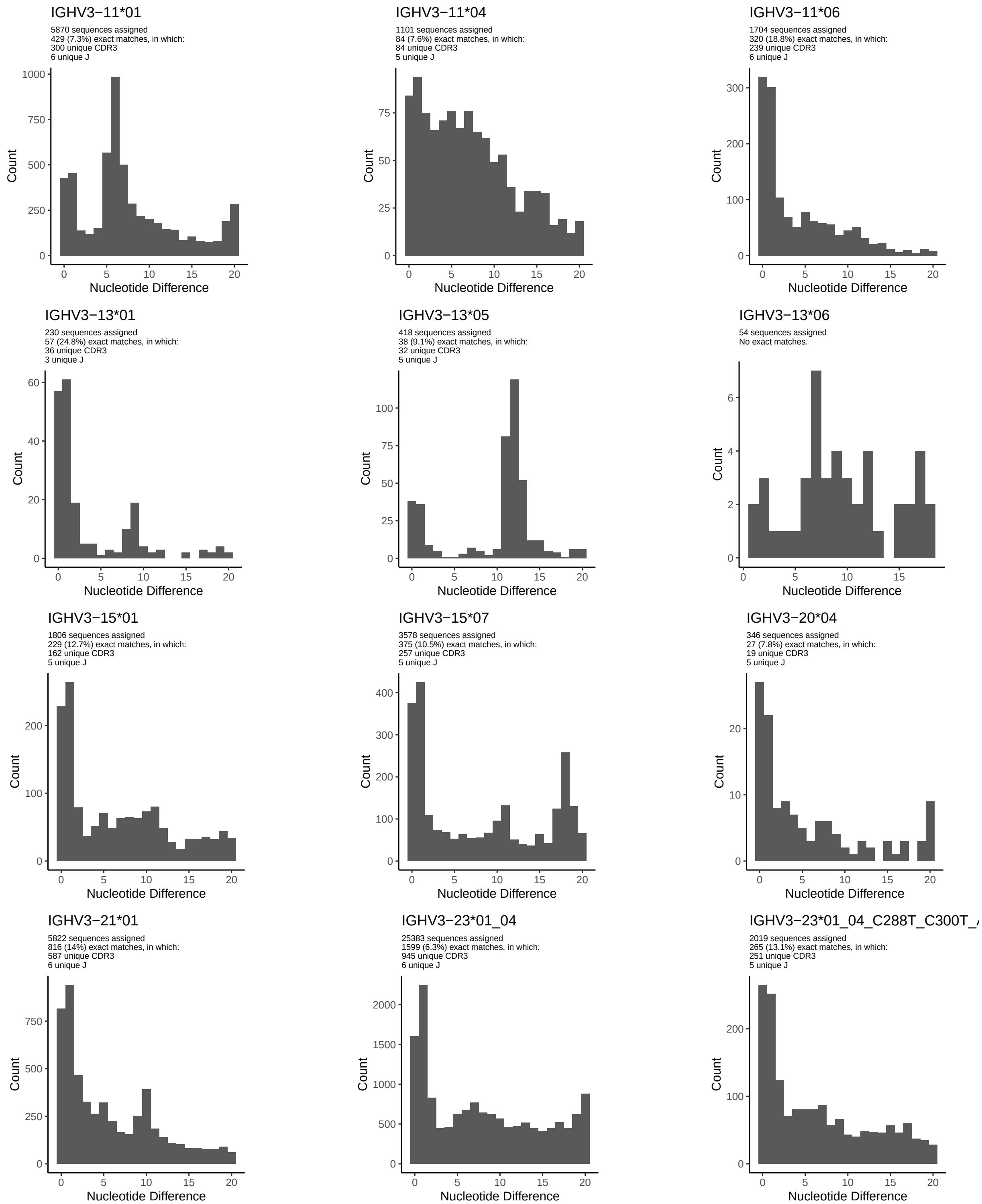

IGHV4−NL1*01_C300T
Count
20
15
10
5
0
7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 24
CDR3 AA Length (unmutated)

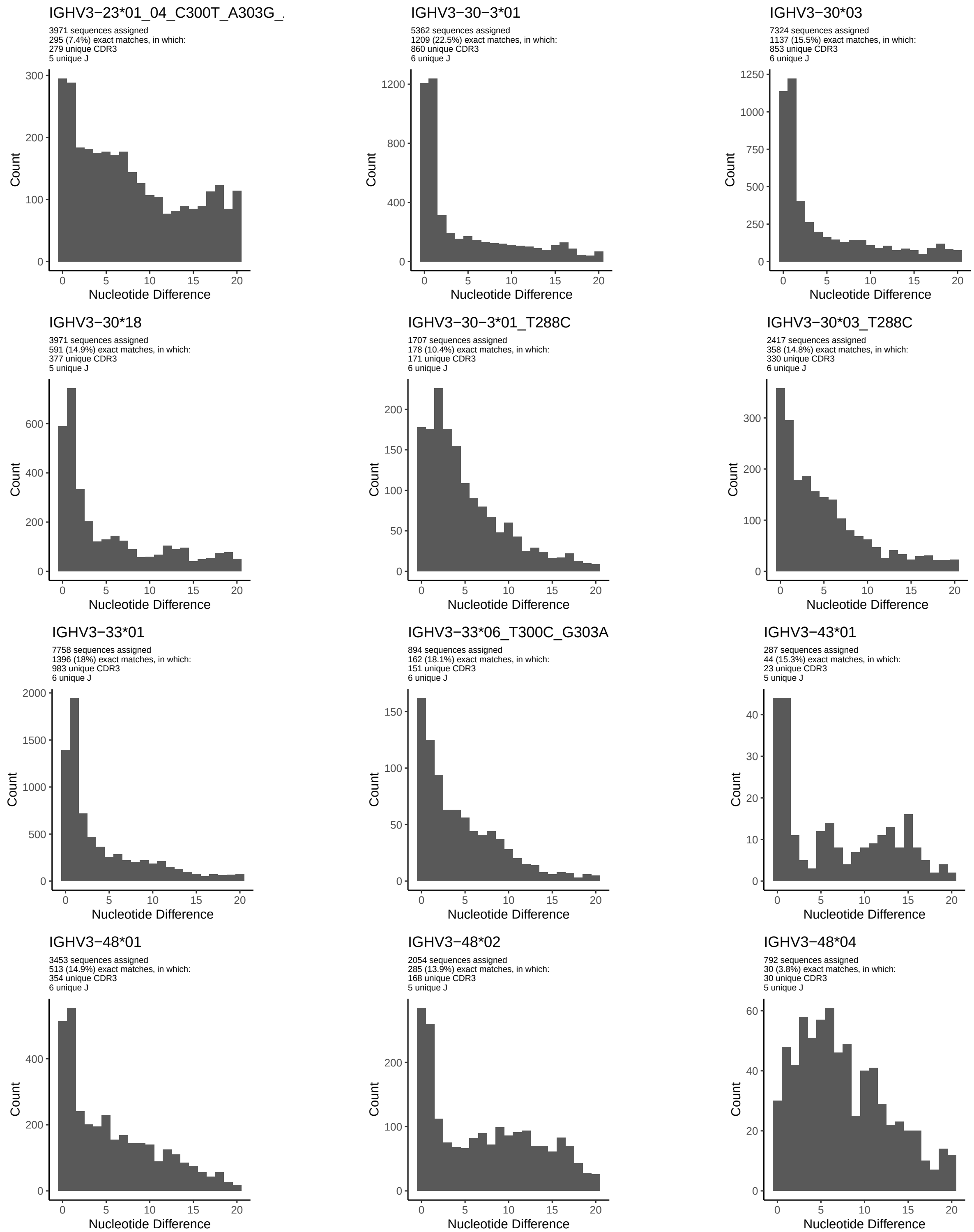
2 Variation from germline, in assignments to each allele



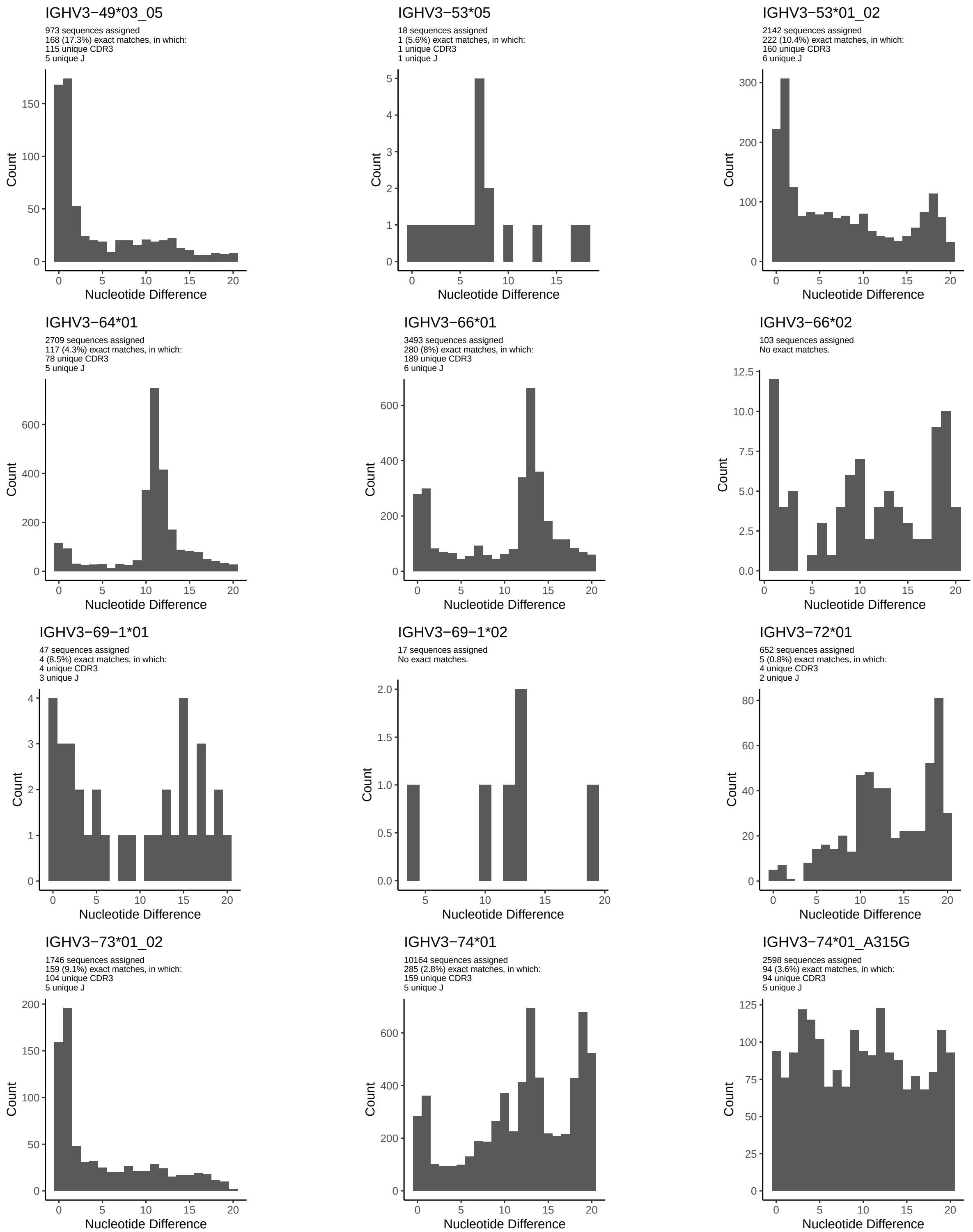


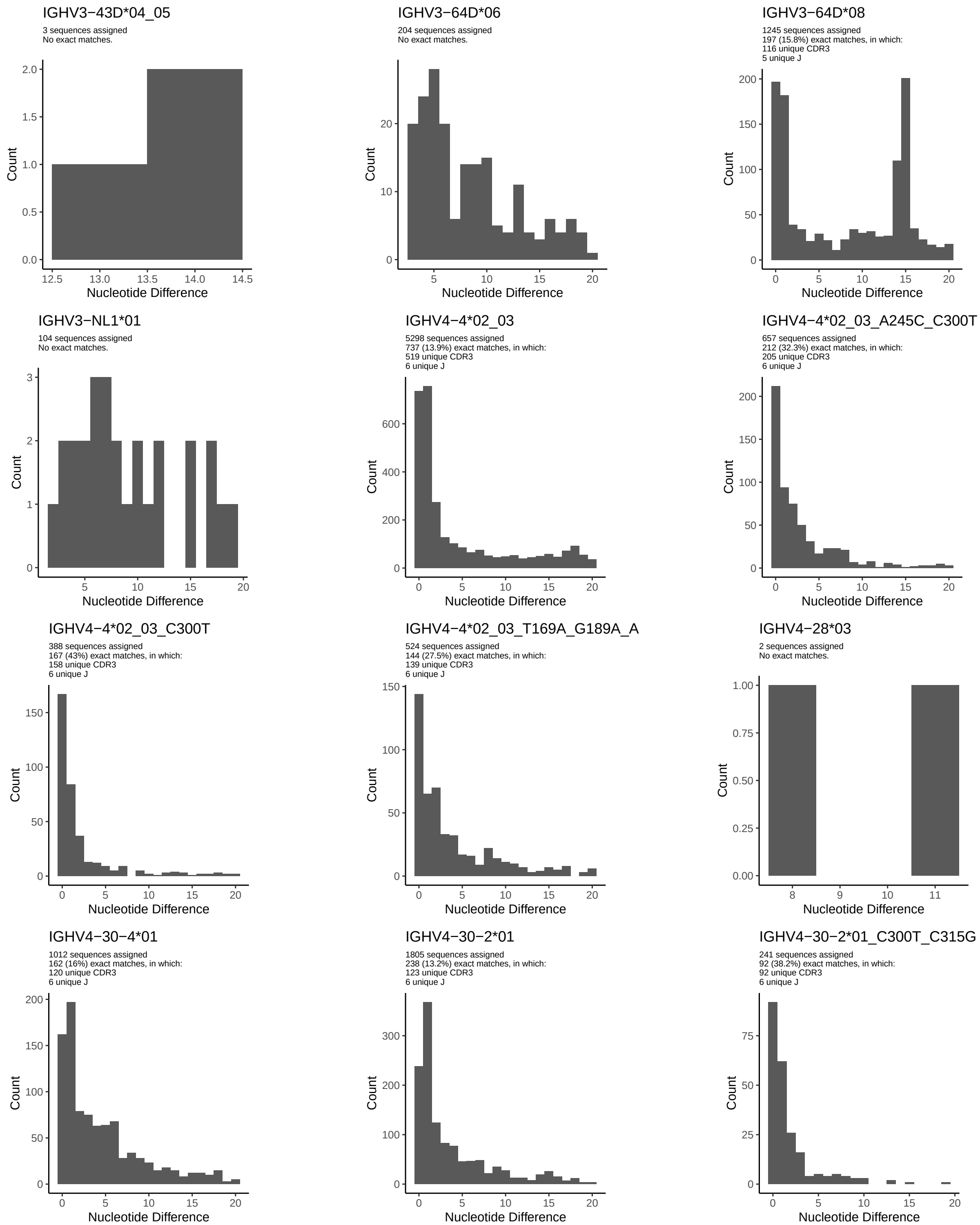
IGHV1−69−2*01
10 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV1−69*01_12_13
6807 sequences assigned
1750 (25.7%) exact matches, in which:
1337 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV1−69*06_14
2931 sequences assigned
749 (25.6%) exact matches, in which:
617 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV2−5*01
372 sequences assigned
47 (12.6%) exact matches, in which:
37 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV2−5*02
1582 sequences assigned
144 (9.1%) exact matches, in which:
98 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−26*01
186 sequences assigned
51 (27.4%) exact matches, in which:
43 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−70*01
386 sequences assigned
78 (20.2%) exact matches, in which:
50 unique CDR3
4 unique J
Count
Nucleotide Difference
IGHV2−70D*04
154 sequences assigned
27 (17.5%) exact matches, in which:
18 unique CDR3
4 unique J
Count
Nucleotide Difference
IGHV3−7*01
11888 sequences assigned
482 (4.1%) exact matches, in which:
354 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−7*03
1015 sequences assigned
38 (3.7%) exact matches, in which:
36 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−7*05
1760 sequences assigned
158 (9%) exact matches, in which:
117 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−9*01
3192 sequences assigned
467 (14.6%) exact matches, in which:
279 unique CDR3
6 unique J
Count
Nucleotide Difference

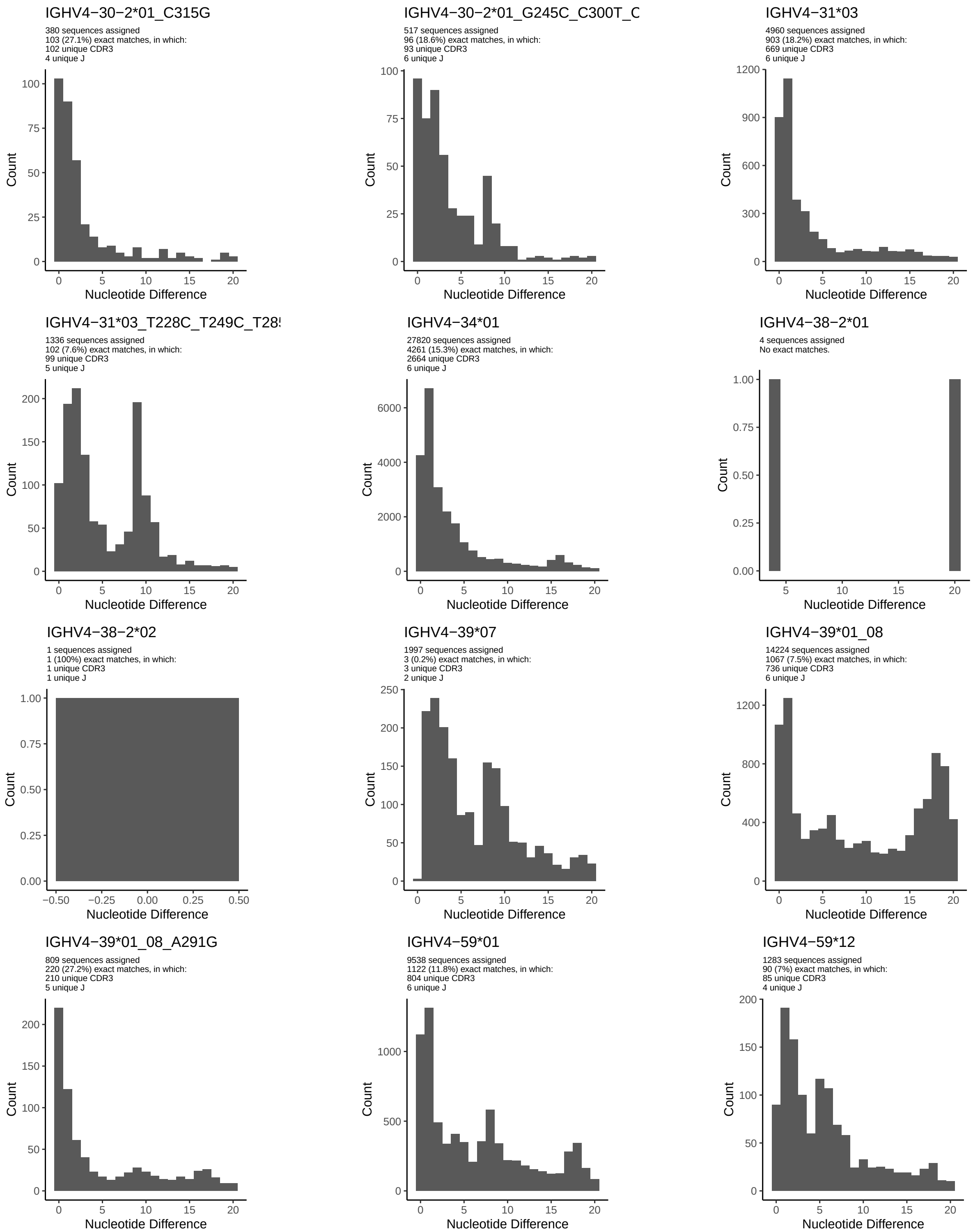

IGHV3−11*01
5870 sequences assigned
429 (7.3%) exact matches, in which:
300 unique CDR3
6 unique J
IGHV3−11*04
1101 sequences assigned
84 (7.6%) exact matches, in which:
84 unique CDR3
5 unique J
IGHV3−11*06
1704 sequences assigned
320 (18.8%) exact matches, in which:
239 unique CDR3
6 unique J
IGHV3−13*01
230 sequences assigned
57 (24.8%) exact matches, in which:
36 unique CDR3
3 unique J
IGHV3−13*05
418 sequences assigned
38 (9.1%) exact matches, in which:
32 unique CDR3
5 unique J
IGHV3−13*06
54 sequences assigned
No exact matches.
IGHV3−15*01
1806 sequences assigned
229 (12.7%) exact matches, in which:
162 unique CDR3
5 unique J
IGHV3−15*07
3578 sequences assigned
375 (10.5%) exact matches, in which:
257 unique CDR3
5 unique J
IGHV3−20*04
346 sequences assigned
27 (7.8%) exact matches, in which:
19 unique CDR3
5 unique J
IGHV3−21*01
5822 sequences assigned
816 (14%) exact matches, in which:
587 unique CDR3
6 unique J
IGHV3−23*01_04
25383 sequences assigned
1599 (6.3%) exact matches, in which:
945 unique CDR3
6 unique J
IGHV3−23*01_04_C288T_C300T_
2019 sequences assigned
265 (13.1%) exact matches, in which:
251 unique CDR3
5 unique J
Count
Nucleotide Difference

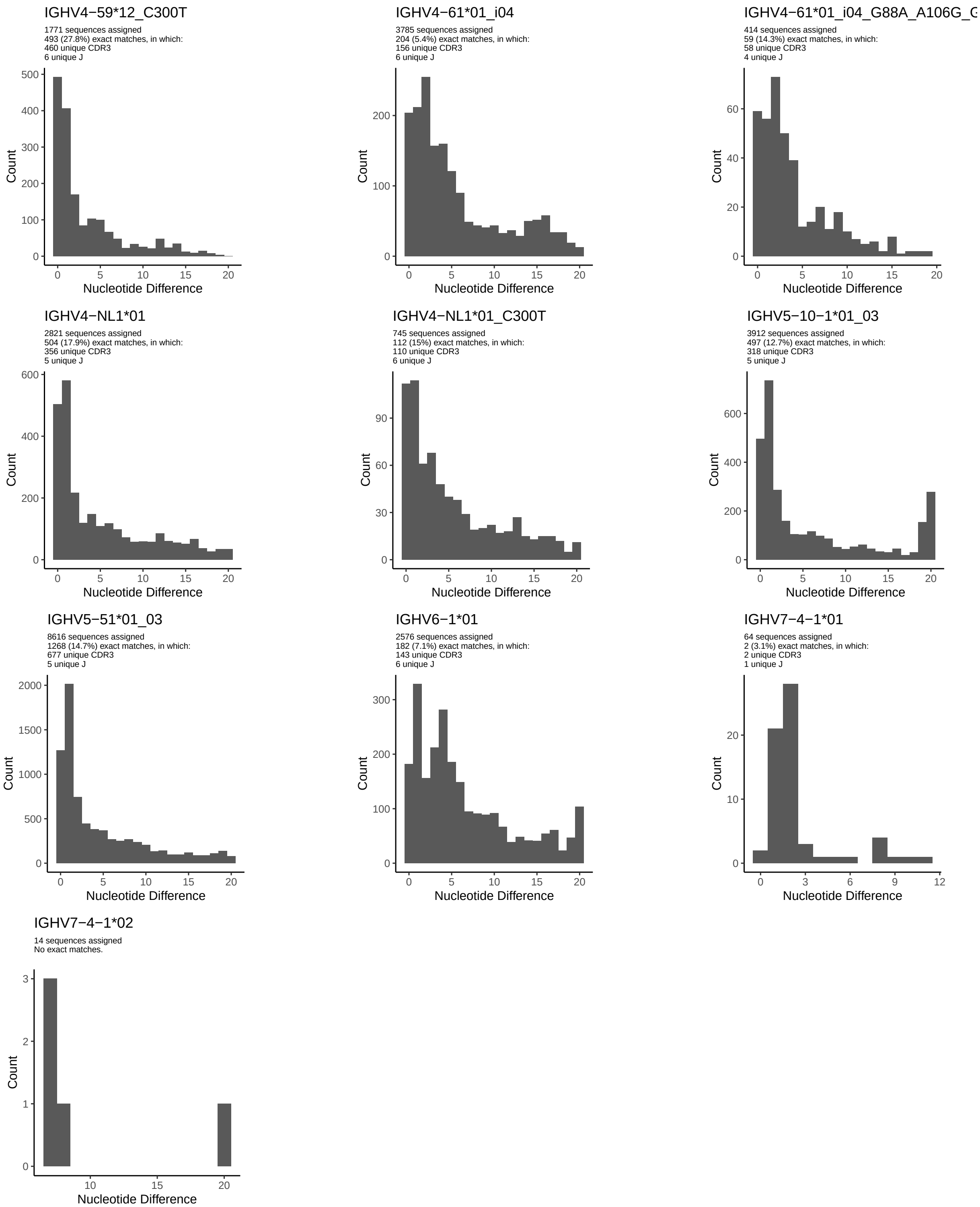


IGHV3−23*01_04_C300T_A303G_
3971 sequences assigned
295 (7.4%) exact matches, in which:
279 unique CDR3
5 unique J
IGHV3−30−3*01
5362 sequences assigned
1209 (22.5%) exact matches, in which:
860 unique CDR3
6 unique J
IGHV3−30*03
7324 sequences assigned
1137 (15.5%) exact matches, in which:
853 unique CDR3
6 unique J
IGHV3−30*18
3971 sequences assigned
591 (14.9%) exact matches, in which:
377 unique CDR3
5 unique J
IGHV3−30−3*01_T288C
1707 sequences assigned
178 (10.4%) exact matches, in which:
171 unique CDR3
6 unique J
IGHV3−30*03_T288C
2417 sequences assigned
358 (14.8%) exact matches, in which:
330 unique CDR3
6 unique J
IGHV3−33*01
7758 sequences assigned
1396 (18%) exact matches, in which:
983 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
894 sequences assigned
162 (18.1%) exact matches, in which:
151 unique CDR3
6 unique J
IGHV3−43*01
287 sequences assigned
44 (15.3%) exact matches, in which:
23 unique CDR3
5 unique J
IGHV3−48*01
3453 sequences assigned
513 (14.9%) exact matches, in which:
354 unique CDR3
6 unique J
IGHV3−48*02
2054 sequences assigned
285 (13.9%) exact matches, in which:
168 unique CDR3
5 unique J
IGHV3−48*04
792 sequences assigned
30 (3.8%) exact matches, in which:
30 unique CDR3
5 unique J
Count
Nucleotide Difference

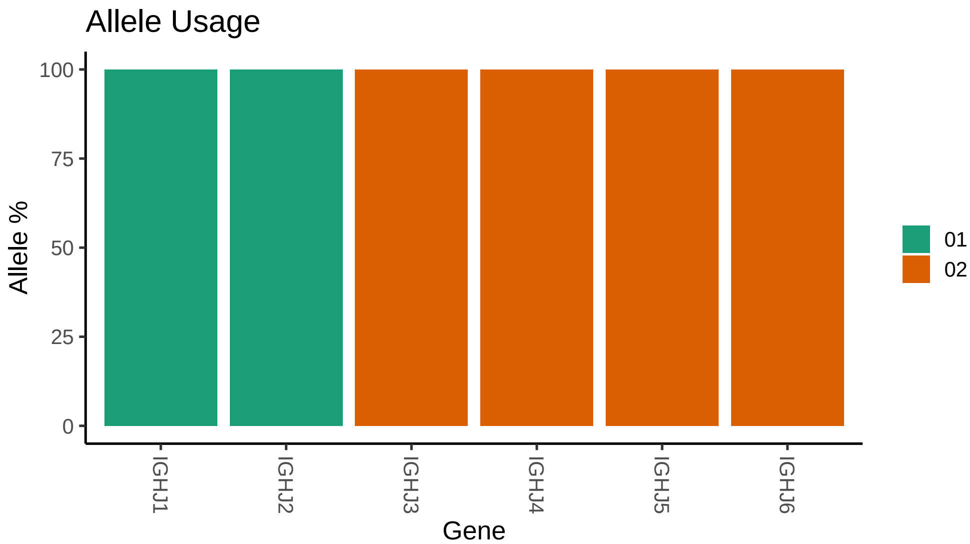

IGHV3−49*03_05
973 sequences assigned
168 (17.3%) exact matches, in which:
115 unique CDR3
5 unique J
IGHV3−53*05
18 sequences assigned
1 (5.6%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−53*01_02
2142 sequences assigned
222 (10.4%) exact matches, in which:
160 unique CDR3
6 unique J
IGHV3−64*01
2709 sequences assigned
117 (4.3%) exact matches, in which:
78 unique CDR3
5 unique J
IGHV3−66*01
3493 sequences assigned
280 (8%) exact matches, in which:
189 unique CDR3
6 unique J
IGHV3−66*02
103 sequences assigned
No exact matches.
IGHV3−69−1*01
47 sequences assigned
4 (8.5%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV3−69−1*02
17 sequences assigned
No exact matches.
IGHV3−72*01
652 sequences assigned
5 (0.8%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV3−73*01_02
1746 sequences assigned
159 (9.1%) exact matches, in which:
104 unique CDR3
5 unique J
IGHV3−74*01
10164 sequences assigned
285 (2.8%) exact matches, in which:
159 unique CDR3
5 unique J
IGHV3−74*01_A315G
2598 sequences assigned
94 (3.6%) exact matches, in which:
94 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV3−43D*04_05
3 sequences assigned
No exact matches.
IGHV3−64D*06
204 sequences assigned
No exact matches.
IGHV3−64D*08
1245 sequences assigned
197 (15.8%) exact matches, in which:
116 unique CDR3
5 unique J
IGHV3−NL1*01
104 sequences assigned
No exact matches.
IGHV4−4*02_03
5298 sequences assigned
737 (13.9%) exact matches, in which:
519 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
657 sequences assigned
212 (32.3%) exact matches, in which:
205 unique CDR3
6 unique J
IGHV4−4*02_03_C300T
388 sequences assigned
167 (43%) exact matches, in which:
158 unique CDR3
6 unique J
IGHV4−4*02_03_T169A_G189A_A
524 sequences assigned
144 (27.5%) exact matches, in which:
139 unique CDR3
6 unique J
IGHV4−28*03
2 sequences assigned
No exact matches.
IGHV4−30−4*01
1012 sequences assigned
162 (16%) exact matches, in which:
120 unique CDR3
6 unique J
IGHV4−30−2*01
1805 sequences assigned
238 (13.2%) exact matches, in which:
123 unique CDR3
6 unique J
IGHV4−30−2*01_C300T_C315G
241 sequences assigned
92 (38.2%) exact matches, in which:
92 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−30−2*01_C315G
380 sequences assigned
103 (27.1%) exact matches, in which:
102 unique CDR3
4 unique J
IGHV4−30−2*01_G245C_C300T_C
517 sequences assigned
96 (18.6%) exact matches, in which:
93 unique CDR3
6 unique J
IGHV4−31*03
4960 sequences assigned
903 (18.2%) exact matches, in which:
669 unique CDR3
6 unique J
IGHV4−31*03_T228C_T249C_T28
1336 sequences assigned
102 (7.6%) exact matches, in which:
99 unique CDR3
5 unique J
IGHV4−34*01
27820 sequences assigned
4261 (15.3%) exact matches, in which:
2664 unique CDR3
6 unique J
IGHV4−38−2*01
4 sequences assigned
No exact matches.
IGHV4−38−2*02
1 sequences assigned
1 (100%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−39*07
1997 sequences assigned
3 (0.2%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV4−39*01_08
14224 sequences assigned
1067 (7.5%) exact matches, in which:
736 unique CDR3
6 unique J
IGHV4−39*01_08_A291G
809 sequences assigned
220 (27.2%) exact matches, in which:
210 unique CDR3
5 unique J
IGHV4−59*01
9538 sequences assigned
1122 (11.8%) exact matches, in which:
804 unique CDR3
6 unique J
IGHV4−59*12
1283 sequences assigned
90 (7%) exact matches, in which:
85 unique CDR3
4 unique J
Count
Nucleotide Difference


IGHV4−59*12_C300T
1771 sequences assigned
493 (27.8%) exact matches, in which:
460 unique CDR3
6 unique J
IGHV4−61*01_i04
3785 sequences assigned
204 (5.4%) exact matches, in which:
156 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G_C
414 sequences assigned
59 (14.3%) exact matches, in which:
58 unique CDR3
4 unique J
IGHV4−NL1*01
2821 sequences assigned
504 (17.9%) exact matches, in which:
356 unique CDR3
5 unique J
IGHV4−NL1*01_C300T
745 sequences assigned
112 (15%) exact matches, in which:
110 unique CDR3
6 unique J
IGHV5−10−1*01_03
3912 sequences assigned
497 (12.7%) exact matches, in which:
318 unique CDR3
5 unique J
IGHV5−51*01_03
8616 sequences assigned
1268 (14.7%) exact matches, in which:
677 unique CDR3
5 unique J
IGHV6−1*01
2576 sequences assigned
182 (7.1%) exact matches, in which:
143 unique CDR3
6 unique J
IGHV7−4−1*01
64 sequences assigned
2 (3.1%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV7−4−1*02
14 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 064/M64 064/M64 064/M64 064/dd/92f06a366e3501dc4
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 064/M64 064/M64 064/M64 064/dd/92f06a366
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 064/M64 064/M64 064/M64 064/dd/92f06a366e3501d
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_T169A_G189A_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T228C_T249C_T285C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 01_08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 01_08_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_G129C_C134A_A147C_T165C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```